Our Universe: Solar System & The Moon(Part - 2)

Introduction

There are eight planets, which revolve around the Sun. Sun is at the centre of the solar system. Some planets in our solar system are called inner and some are outer planets. Inner and outer planets are categorized according to the distance from the Sun. In this chapter, we will study about the solar system and its components.

Our Universe

Universe is everything that exists in totality. For example stars, planets, Sun, steroids, all other celestial bodies etc. All stars and clouds are made up of hydrogen and helium.

Fig: Universe

Astronomy

The branch of science that deals with the study of space Is called astronomy.

Planets

Planet is a celestial body made up of rocks, metals and gases, which revolve around a star, for example Earth, Mars, Jupiter etc. There are 8 planets in our solar system.

Star

Star is a ball of hot and bright gases.

There are billions of stars in the universe. Stars appear to twinkle from the surface of the earth.

Solar System

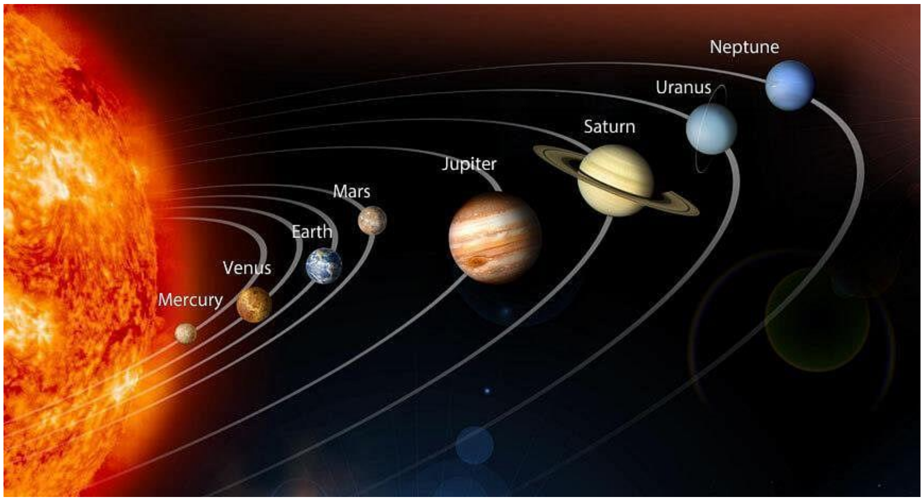


Fig: Solar System

Solar system means the family of the Sun that is the family of 8 planets orbiting around the Sun. It also includes the moons of the planets, asteroids, meteors and comets etc.

Outer planets: The inner and outer planets are differentiated according to their distance from the Sun. There are four outer planets in our solar system, these are, Jupiter, Saturn, Uranus and Neptune.

Inner planets: There are four inner planets, which are closest to the sun and revolve around the sun. These are Mercury, Venus, Earth and Mars.

Fig: Inner Orbit Planets

Sun

Sun is a medium sized, middle-aged star and a ball of bright and hot gases. It has been burning and shining since billions of years and would continue to do so for about 10000000000 years.

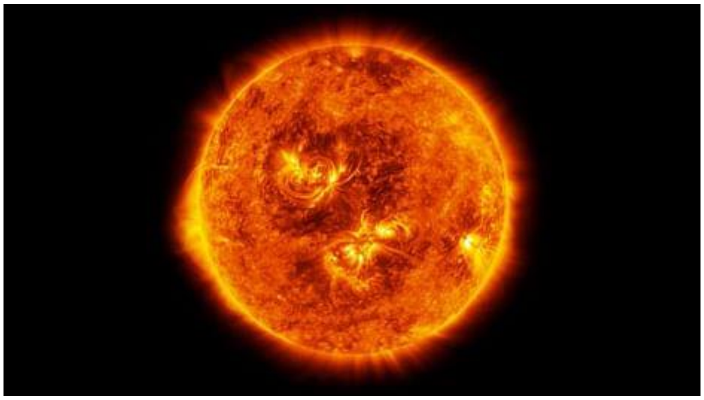


Fig: The Sun

The temperature of the Sun's surface is about 6'000°c and the hotness increases from the surface to the core of the Sun. The Sun has spots, which are little bit cooler than the rest part. The Sun is in the second stage of its career. It is the nearest star to the Earth, about 15 crore kilometer from us. The diameter of Sun is 110 time the earth's diameter. The Sun's light looks white but actually it's a mixture o seven colors.

Violet
Indigo
Blue
Green
Yellow
Orange
Red

Solar Eclipse

This phenomenon occurs when moon (Which circles around the earth) passes directly between the earth and the sun and hides the sun partially or full, from Earth's view.

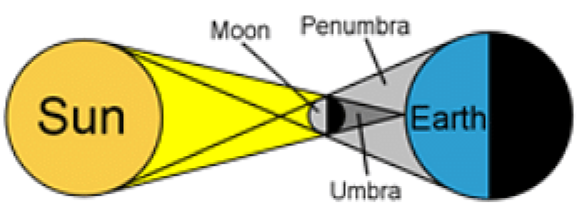


Fig: Solar Eclipse

Solar eclipse can only happen during a new moon. Two to five solar eclipses occur in a year.

Planets

We have studied about the planets of our solar system in previous classes. We will study only these four planets here.

Mercury

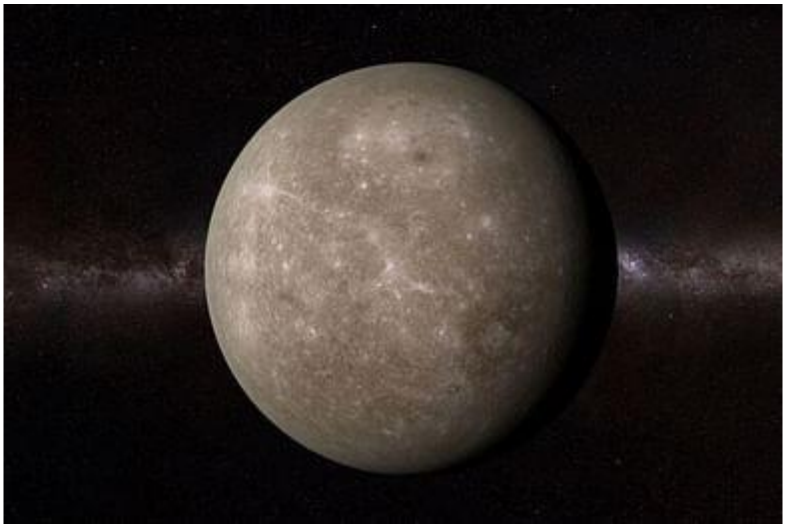


Fig: Mercury

It is named after the name of the messenger of the Roman gods. Mercury takes 58 days to spin on its axis and 88 days to revolve around the Sun. There is no atmosphere and water on this planet. It is covered in craters. The biggest craters on this planet is calories basin.

Venus

Fig: Venus

It is named after the Roman goddess of love and beauty. It is the most beautiful object in the sky. This planet was first observed by Galileo. Venus is the brightest and hottest planet, because it has atmosphere to trap the sun's light making it hottest, it is called as the morning and evening star. Venus rotates on its axis very slowly but it revolves around the sun fast and hence 1 day on Venus = 243 earth days.

But 1 year on Venus = 225 earth days. It has no moon.

Earth

Fig: The Earth

The only planet in the solar system that has life. It has all the essential components of life i.e.

Appropriate warmth from the Sun;

Water, about 71 % of the surface is covered with water; Oxygen

The first life on the earth appeared about 3 billions years ago. It is also named after Roman goddess "Terra" meaning earth. The temperature of the earth's surface is 60°c and it increases as one goes deeper and deeper.

Earth is magnetic because of the presence of molten iron in the core. The needle of the compass always points towards the magnetic north pole.

The earth rotates on its axis in approximately 23 hours, 56 minutes and 22.7 seconds. It revolves around the sun in 365 days, 6 hours, 9 minutes and 10 seconds.

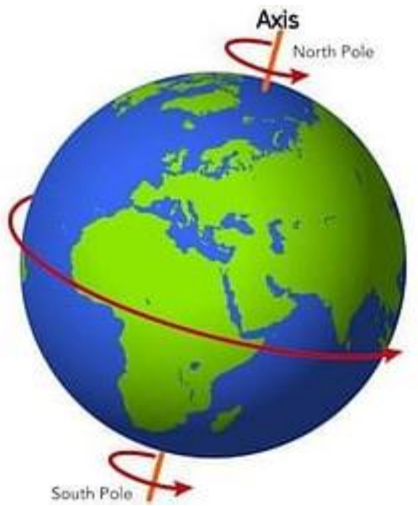


Fig: Rotation of earth around its own axis

Object on this planet doesn't fly off because of the force of gravity, which pulls every object towards itself.

Gravitation

Every planet in the universe attracts each other with a force proportional to their mass. The earth attracts the object when thrown upwards due to the force of gravity. The force of attraction of the earth is due to the structure of the earth's core.

Why does all objects fall towards the earth instead of going up?

Why do we feel being pulled down when we jump?

Why it is easier to fail than to jump?

The answer lies in this fact that there is something in the earth which tries all the time to pull all the objects towards itself-

This something in the Earth is called as gravity or gravitational force. Gravitation is a force with which every objects fall towards the surface of the earth.

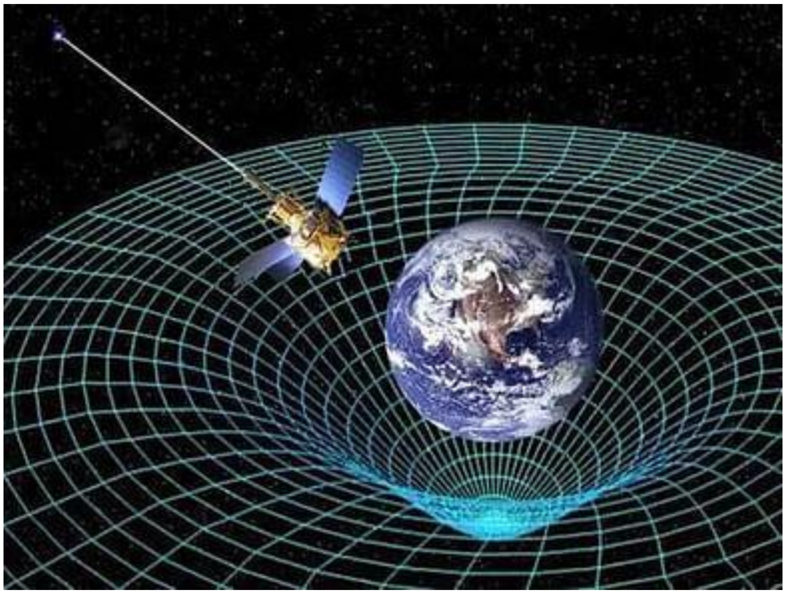


Fig: Gravitation

When we jump we do work against gravity, hence force is applied and energy is spent.

When anything falls, it falls with gravity hence little force is applied and little energy is consumed.

How the concept of Gravity came into light

Fig: Isaac Newton sitting under the tree (Discovery of Universal law of gravitation)
One day sir Isaac Newton was sitting under an apple tree, an apple fell on his head. He started thinking "Why does it happen, why all the objects fall towards ground only". Suddenly he got an idea about gravity and formulated universal law of gravitation. He has a brilliant insight that answered many questions.

Mass and Weight

We have already discussed about the mass, now in this chapter we will discuss about weight. Weight can be defined as the product of force with which an object is attracted towards the earth and mass of the object. The unit of weight is kilograms and unit of force is new ton. The force of attraction with which the earth attracts the object is called gravitational force.
Mass is a constant quantity. Mass remains same even if we are in space, but our weight can become zero in space or at some places.
People feel weightless in the space and hence float in the space.

Look at the following picture of astronaut floating in the space:

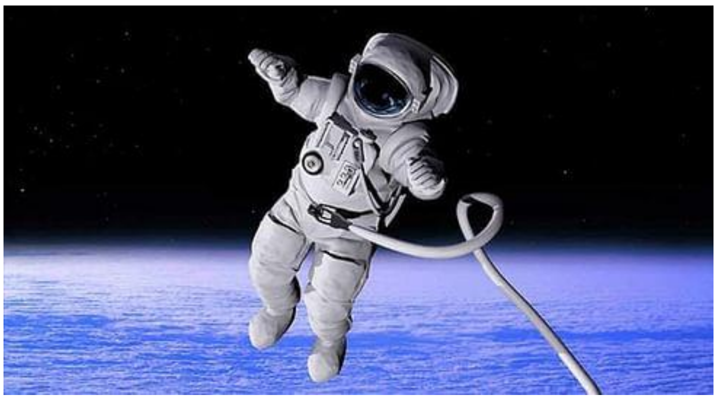


Fig: An astronaut floating in the space

Astronaut feeling weightlessness in the sky

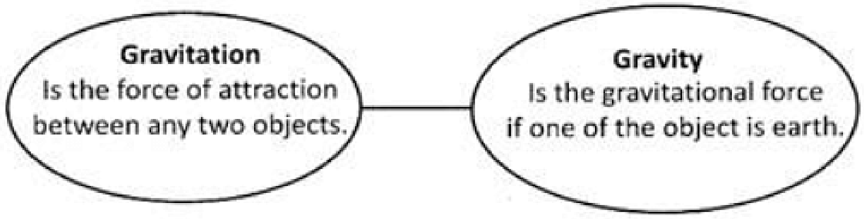
Weight = Mass x acceleration due to gravity
W = M x g
Value of g = 9.8 on the earth's surface.
Value of g = 9.8 / 6 on the surface of the moon.

A boy weighing 24 kg on the earth would weigh 4 kg on the moon. The weight on the moon reduces to 1/6th times that on the earth.

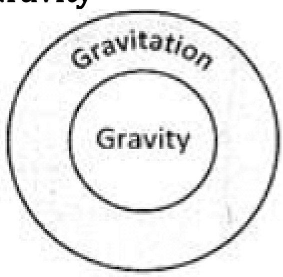
Sun has strong force of gravity that holds all the eight planets and other heavenly bodies like asteroid of our solar system.

Movement of moons around their planets and movement of satellites and spacecrafts around the earth is due to this force.

Relation between Gravitation and Gravity



Gravity



The force of gravity decreases as the distance between the two objects increases.

Newton's Universal Law of Gravitation

Newton formulated a law about gravitation that holds the force between two bodies. Hence this law is called as universal law of gravitation.

This law states that

Every two object in the universe attract each other by a force which is directly proportional to the product of their masses and inversely proportional to the

F is directly proportional to the product of the two masses ,i.e m1 * m2

F is inversely proportional to the distance between them .i.e $\frac{1}{r^2}$

M_1 and M_2 - masses of the two objects

r - distance between them

G- Gravitation constant = 6.674×10^{-11} Nm^2/kg^2

F - Force between the masses

$$F = G\frac{M_1 \times M_2}{r^2}$$

The Sl unit of force is newton and mass is measured in kilograms and distance between The objects is measured in metre. The value of acceleration due to gravity on the surface of the earth is approximately $9.81 m/s^2$.

Moon

Fig: The Moon

Planets revolve around the Sun, similarly some heavenly bodies revolve around these planets, such as Moon revolves around the Earth.

Earth has one moon called as Luna. It looks beautiful and bright in the night sky. It has no light of its own, shines by reflected light of the Sun. Moon changes its shape from night to night and it completes one cycle in a month. The changes in the shape of Moon are called phases of Moon.

The phases of Moon are explained below:

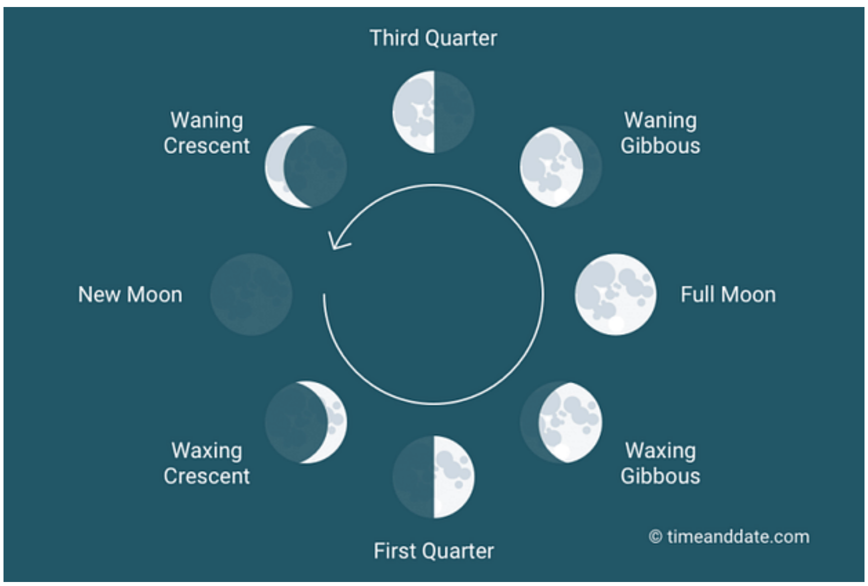


Fig: Various phases of moon

Crescent moon in the sky

Gibbous moon